

A candidate proof of Informal Conjecture on Property (*) for $d = o(n/\log n)$

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

A half-interval decomposition of the vertex labels, followed by read- d Hölder bounds for uniform spacings, proves Property (*) throughout $d = o(n/\log n)$. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Informal Conjecture on Property \(*\) for \$d = o\(n/\log n\)\$](#) from *On random irregular subgraphs*.

1.1 Problem statement

The campaign artifact does not contain a separately extracted statement; see the catalog entry linked above.

1.2 Claimed result

Model verdict: **proved** (confidence: high).

A half-interval decomposition of the vertex labels, followed by read- d Hölder bounds for uniform spacings, proves Property (*) throughout $d = o(n/\log n)$.

The model's own caveat: Literature priority has not been independently checked; graphs are finite, simple, and undirected.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 Property (*) in the conjectured range

2.1.1 1. Statement and the key observation

Let G be a d -regular graph on n vertices. Independently assign $X_v \sim \text{Unif}[0, 1]$, and retain $uv \in E(G)$ exactly when $X_u + X_v \geq 1$. Write

$$m_k = |\{v : \deg_H(v) = k\}|, \quad \mu = \frac{n}{d+1}.$$

We prove the following.

2.1.1.1 Theorem For every sequence of d -regular graphs with

$$d = o(n/\log n),$$

with probability tending to 1,

$$\max_{0 \leq k \leq d} |m_k - \mu| = o(\mu).$$

The probability estimates are uniform over the choice of G .

More quantitatively, there is an absolute constant C such that, for all sufficiently large n , whenever $d \geq 64 \log n$,

$$\Pr \left(\max_{0 \leq k \leq d} |m_k - \mu| > C \left(\sqrt{\frac{n}{d} \log n} + \log n \right) \right) \leq Cn^{-3}. \quad (1)$$

The useful idea is to record whether each label belongs to the lower or upper half of $[0, 1]$. Conditional on these bits:

- every lower–lower edge is absent;
- every upper–upper edge is present;
- only edges across the resulting bipartition remain random.

Thus one can use the independent-side argument from the supplied earlier attempt **without properly coloring** G . The additional random conditional mean is a sum of bounded local functions of independent bits, and a read- $(d + 1)$ Hölder inequality controls it.

We verify all these steps below.

2.1.2 2. Two elementary tools

2.1.2.1 2.1. Product-space Hölder inequality Suppose $Z_1, \dots, Z_N$ are independent, and each nonnegative function f_i depends on some subset of these coordinates. If every coordinate occurs in at most R functions, then

$$\mathbb{E} \prod_i f_i \leq \prod_i (\mathbb{E} f_i^R)^{1/R}. \quad (2)$$

For completeness, integrate one coordinate at a time. At a coordinate occurring in $q \leq R$ factors, apply Hölder with exponent R to those q factors, adding a constant factor when $q < R$. The resulting factors have the form

$$(\mathbb{E}_{Z_j} f_i^R)^{1/R}.$$

Repeating over the remaining coordinates gives (2).

In particular, if Y_i are functions with this read- R property, then

$$\log \mathbb{E} e^{\lambda \sum_i (Y_i - \mathbb{E} Y_i)} \leq \frac{1}{R} \sum_i \log \mathbb{E} e^{\lambda R (Y_i - \mathbb{E} Y_i)}. \quad (3)$$

2.1.2.2 2.2. Uniform spacings Let B_r be any specified spacing between r independent uniform points in $[0, 1]$, including the two endpoint spacings. Then

$$B_r \sim \text{Beta}(1, r), \quad \mathbb{E}B_r = \frac{1}{r+1},$$

and, for every integer $j \geq 1$,

$$\mathbb{E}B_r^j = \frac{j!}{(r+1)(r+2)\cdots(r+j)}. \quad (4)$$

These identities follow directly from the constant density of the spacing vector on the simplex.

If $r \geq d/4$, then

$$\mathbb{E}(dB_r)^j \leq 4^j j!.$$

Consequently, for $|\theta| \leq 1/8$, expansion of the exponential gives

$$\begin{aligned} \log \mathbb{E}e^{\theta dB_r} &\leq \mathbb{E}e^{\theta dB_r} - 1 \\ &\leq \theta \frac{d}{r+1} + \sum_{j \geq 2} (4|\theta|)^j \\ &\leq \theta \frac{d}{r+1} + 32\theta^2. \end{aligned} \quad (5)$$

The same bound holds for a variable identically zero, with its mean replacing $d/(r+1)$.

2.1.3 3. Exposing the half-interval bits

Generate the labels as

$$X_v = \frac{C_v + U_v}{2},$$

where all $C_v \sim \text{Bernoulli}(1/2)$ and $U_v \sim \text{Unif}[0, 1]$ are mutually independent.

Put

$$V_a = \{v : C_v = a\}, \quad a \in \{0, 1\}.$$

For each vertex, define its degree across this random cut by

$$r_v = |\{u \in N_G(v) : C_u \neq C_v\}|,$$

and define

$$b_v = \begin{cases} 0, & C_v = 0, \\ d - r_v, & C_v = 1. \end{cases}$$

Almost surely, the degree in H is

$$\deg_H(v) = b_v + \sum_{\substack{u \in N_G(v) \\ C_u \neq C_v}} \mathbf{1}_{\{U_u + U_v \geq 1\}}. \quad (6)$$

Indeed, monochromatic edges are deterministic after exposing the bits, while a cross-edge is retained exactly when $U_u + U_v \geq 1$.

For a fixed bit configuration, let

$$q_{v,k}(C) = \frac{\mathbf{1}_{\{b_v \leq k \leq b_v + r_v\}}}{r_v + 1}, \quad Q_k(C) = \sum_v q_{v,k}(C). \quad (7)$$

The spacing identity, or integration of a binomial distribution, shows that

$$q_{v,k}(C) = \Pr(\deg_H(v) = k \mid C).$$

Thus

$$Q_k(C) = \mathbb{E}[m_k \mid C]. \quad (8)$$

We will use the event

$$\mathcal{E} = \{r_v \geq d/4 \text{ for every } v\}.$$

Conditional on either value of C_v , the variable r_v is $\text{Bin}(d, 1/2)$. The elementary binomial lower-tail bound therefore gives

$$\Pr(r_v < d/4) \leq e^{-d/16}, \quad \Pr(\mathcal{E}^c) \leq ne^{-d/16}. \quad (9)$$

2.1.4 4. Concentration conditional on the bits

Fix a bit configuration belonging to $\mathcal{E}$, and fix k . Define

$$M_{a,k} = \sum_{v \in V_a} \mathbf{1}_{\{\deg_H(v)=k\}}, \quad Q_{a,k} = \sum_{v \in V_a} q_{v,k}.$$

Thus $m_k = M_{0,k} + M_{1,k}$ and $Q_k = Q_{0,k} + Q_{1,k}$.

We first bound the moment-generating function of $M_{a,k}$.

Condition additionally on all U_u with $u \in V_{1-a}$. The degree indicators for vertices in V_a are then independent: by (6), each is a function only of its own remaining label U_v .

Let their conditional success probabilities be $p_{v,k}$. If

$$0 \leq k - b_v \leq r_v,$$

then $p_{v,k}$ is the corresponding spacing among the r_v points

$$(1 - U_u)_{\substack{u \in N_G(v) \\ C_u \neq C_v}}.$$

Otherwise $p_{v,k} = 0$. Consequently,

$$\mathbb{E}[p_{v,k} \mid C] = q_{v,k}, \quad (10)$$

and, in the nonzero case, its marginal conditional distribution is $\text{Beta}(1, r_v)$.

For real λ , set $\eta = e^\lambda - 1$. Conditional independence gives

$$\begin{aligned} \mathbb{E}[e^{\lambda M_{a,k}} \mid C] &= \mathbb{E}_{U_{V_{1-a}}} \prod_{v \in V_a} (1 + \eta p_{v,k}) \\ &\leq \mathbb{E}_{U_{V_{1-a}}} \exp \left(\eta \sum_{v \in V_a} p_{v,k} \right). \end{aligned} \quad (11)$$

This inequality is valid for either sign of λ , since $\eta > -1$.

Each opposite-side label is used in at most d of the functions $p_{v,k}$. Applying (2) with $R = d$, and then (5), yields, for $|\eta| \leq 1/8$,

$$\begin{aligned} \log \mathbb{E}[e^{\lambda M_{a,k}} \mid C] &\leq \frac{1}{d} \sum_{v \in V_a} \log \mathbb{E}[e^{\eta d p_{v,k}} \mid C] \\ &\leq \eta Q_{a,k} + 32 \frac{|V_a|}{d} \eta^2. \end{aligned} \quad (12)$$

On $\mathcal{E}$,

$$Q_{a,k} \leq \frac{4|V_a|}{d}.$$

Since $e^\lambda - 1 - \lambda = O(\lambda^2)$, equation (12) implies that, for absolute constants $C_0, \lambda_0 > 0$,

$$\log \mathbb{E}[e^{\lambda(M_{a,k} - Q_{a,k})} \mid C] \leq C_0 \frac{|V_a|}{d} \lambda^2, \quad |\lambda| \leq \lambda_0. \quad (13)$$

The two sides need not be independent. Cauchy–Schwarz nevertheless gives

$$\begin{aligned} \mathbb{E}[e^{\lambda(m_k - Q_k)} \mid C] &\leq (\mathbb{E}[e^{2\lambda(M_{0,k} - Q_{0,k})} \mid C])^{1/2} \\ &\quad \times (\mathbb{E}[e^{2\lambda(M_{1,k} - Q_{1,k})} \mid C])^{1/2}. \end{aligned}$$

Hence, after adjusting absolute constants,

$$\log \mathbb{E}[e^{\lambda(m_k - Q_k)} \mid C] \leq C_1 \frac{n}{d} \lambda^2, \quad |\lambda| \leq \lambda_1. \quad (14)$$

Chernoff optimization proves the following estimate, uniformly over every bit configuration in $\mathcal{E}$:

$$\Pr(|m_k - Q_k| \geq t \mid C) \leq 2 \exp \left[-c \min \left\{ \frac{dt^2}{n}, t \right\} \right]. \quad (15)$$

This handles both tails and all degree values, including $k = 0, d$.

2.1.5 5. Concentration of the random conditional mean

We next control $Q_k(C)$. It is important here **not** to condition the independent bits on $\mathcal{E}$.

Instead, define local truncations

$$\bar{q}_{v,k} = q_{v,k} \mathbf{1}_{\{r_v \geq d/4\}}, \quad \bar{Q}_k = \sum_v \bar{q}_{v,k}.$$

For every bit configuration,

$$0 \leq \bar{q}_{v,k} \leq \frac{4}{d}. \quad (16)$$

Moreover, $\bar{q}_{v,k}$ depends only on the bits in $N_G[v]$. Each bit belongs to exactly $d + 1$ such closed neighborhoods.

Apply (3) with $R = d + 1$, followed by the elementary Hoeffding moment-generating-function bound for a variable in an interval of length $4/d$. For every real λ ,

$$\begin{aligned} \log \mathbb{E} e^{\lambda(\bar{Q}_k - \mathbb{E} \bar{Q}_k)} &\leq \frac{1}{d+1} \sum_v \log \mathbb{E} e^{\lambda(d+1)(\bar{q}_{v,k} - \mathbb{E} \bar{q}_{v,k})} \\ &\leq \frac{n(d+1)}{8} \left(\frac{4}{d} \right)^2 \lambda^2 \\ &\leq 4 \frac{n}{d} \lambda^2. \end{aligned}$$

Therefore

$$\Pr(|\bar{Q}_k - \mathbb{E} \bar{Q}_k| \geq t) \leq 2 \exp \left(-\frac{dt^2}{16n} \right). \quad (17)$$

We also need its mean. Conditional on $X_v = x$, the original degree is $\text{Bin}(d, x)$, so

$$\Pr(\deg_H(v) = k) = \int_0^1 \binom{d}{k} x^k (1-x)^{d-k} dx = \frac{1}{d+1}. \quad (18)$$

Thus $\mathbb{E}m_k = \mathbb{E}Q_k = \mu$. Since $0 \leq q_{v,k} \leq 1$, equations (9) and (18) give

$$0 \leq \mu - \mathbb{E}\bar{Q}_k = \sum_v \mathbb{E}[q_{v,k} \mathbf{1}_{\{r_v < d/4\}}] \leq ne^{-d/16}. \quad (19)$$

Finally, on $\mathcal{E}$,

$$Q_k = \bar{Q}_k \quad \text{for every } k. \quad (20)$$

2.1.6 6. Simultaneous concentration for $d \geq 64 \log n$

Put $B = ne^{-d/16}$. Combining (15), (17), (19), and (20), and taking a union bound over the $d+1$ degree values, gives

$$\Pr \left(\max_{0 \leq k \leq d} |m_k - \mu| \geq 2t + B \right) \leq B + 4(d+1) \exp \left[-c \min \left\{ \frac{dt^2}{n}, t \right\} \right]. \quad (21)$$

Here the bad event $\mathcal{E}^c$, which is common to all k , is counted only once. This inequality is valid for every $d \geq 1$, although it is useful in this form when d is at least a sufficiently large multiple of $\log n$.

Suppose now that $d \geq 64 \log n$. Then

$$B \leq n^{-3}.$$

Choose

$$t = A \left(\sqrt{\frac{n}{d} \log n} + \log n \right),$$

where $A \geq 1$ is a sufficiently large absolute constant. We have

$$\frac{dt^2}{n} \geq A^2 \log n, \quad t \geq A \log n.$$

Thus (21) proves (1).

Under the conjectured hypothesis $d = o(n/\log n)$,

$$\frac{\sqrt{(n/d) \log n} + \log n}{n/(d+1)} = O \left(\sqrt{\frac{d \log n}{n}} + \frac{d \log n}{n} \right) = o(1). \quad (22)$$

This proves Property (*) in the large-degree case.

2.1.7 7. The remaining small degrees

For completeness, we handle all $d < 64 \log n$ directly.

For fixed k , let

$$I_{v,k} = \mathbf{1}_{\{\deg_H(v)=k\}}.$$

Each indicator depends on the independent labels in $N_G[v]$, so these functions have the read- R property with

$$R = d + 1.$$

Each indicator is Bernoulli with mean $p = 1/(d + 1)$, by (18). Hence (2) gives

$$\mathbb{E}e^{\lambda m_k} \leq (1 - p + pe^{\lambda R})^{n/R}.$$

The usual elementary Chernoff calculation therefore yields, for $0 < \varepsilon \leq 1$,

$$\Pr(|m_k - \mu| \geq \varepsilon \mu) \leq 2 \exp\left(-\frac{\varepsilon^2 n}{3(d+1)^2}\right). \quad (23)$$

Taking a union bound,

$$\Pr\left(\max_k |m_k - \mu| \geq \varepsilon \mu\right) \leq 2(d+1) \exp\left(-\frac{\varepsilon^2 n}{3(d+1)^2}\right). \quad (24)$$

For $d < 64 \log n$, choose

$$\varepsilon_n = 4(d+1) \sqrt{\frac{\log n}{n}}.$$

Uniformly throughout this degree range, $\varepsilon_n = o(1)$, and (24) tends to zero. The case $d = 0$ is deterministic: $m_0 = n$.

Together with Section 6, this covers every sequence $d = o(n/\log n)$, completing the proof. $\square$

2.1.8 8. Scope and relation to the earlier obstruction

The supplied attempt's read- R argument is valid, but a proper coloring of G is unnecessary. Exposing the half-interval bit of every label makes the **random part** of the model bipartite, regardless of $\chi_f(G)$. The monochromatic edges contribute only the deterministic offsets b_v .

The remaining conditional mean is controlled because, outside exponentially unlikely local cut imbalances, each of its summands is at most $4/d$ and reads only $d + 1$ independent bits. This removes the chromatic-number loss and avoids the unresolved residual estimate in the earlier approach.

There is no remaining mathematical gap for the stated simple-graph model. The argument does not establish failure beyond the conjectured range, and literature priority remains unchecked.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2207.13651__00
Catalog entry	Informal Conjecture on Property (*) for d = o(n/log n)
Source corpus	arxiv
Campaign leg	selected second attempts (<code>attacks_retry</code>)
Source paper / entry	On random irregular subgraphs
Model verdict	proved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>
Model	<code>gpt-6-astra</code> , reasoning effort max, mode= <code>pro</code> , flex service tier

Original artifact	attacks_retry/2207.13651__00/output.md
Independent review	None
arXiv	arXiv:2207.13651
Previous attempt	attacks/2207.13651__00: gpt-5.6-sol verdict partial
Model's caveats	Literature priority has not been independently checked; graphs are finite, simple, and undirected.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/2207.13651__00/`.